MAD Framework Architecture

Dual-Agent Adversarial Critique Project

Practical Multi-Agent Debate (MAD) Implementation

Overview

This is a practical, evolving implementation of **Multi-Agent Debate (MAD)** — the research approach in which multiple LLM agents propose, critique, and refine answers to improve reliability, completeness, and factuality (building on Irving et al. 2018 AI Debate and Du et al. 2023/ICML 2024).

Core Agents and Roles

Agent	Full Name	Primary Mandate	Key Output
AA	Advocate Agent	Build the strongest, most complete case	Thorough positive analysis, evidence, options
DA	Devil's Advocate Agent	Attack completeness, surface gaps, risks, unasked questions, failure modes	Structured critiques and pressure tests
SSA	Synthesizer Agent	Integrate AA + DA into the closest available tested result while preserving residual uncertainty	Final synthesis + executive summary + next-best alternatives + horizon analysis
SA	Super Agent	Meta-researcher: track latest MAD methods, update protocols, maintain living knowledge base	Prompt/architecture improvements and checklists fed back into AA/DA/SSA

Execution Flow

- Topic / query enters the system.
- **AA** produces a strong affirmative case.
- **DA** critiques AA's output (and the topic) rigorously.
- **SSA** synthesizes, separates high-confidence vs contested claims, surfaces residual uncertainties, applies relevant checklists (e.g., Device SWAT Arsenal), and always closes with next-best alternative recommendations (single-item or dual-comparison) plus horizon analysis.
- **SA** operates asynchronously (weekly automation + on-demand): researches new MAD techniques, extracts failure modes from completed runs, and evolves the prompts, checklists, and protocols.

Supporting Elements

- **Living knowledge base**: Topic-specific checklists (e.g., MAD_Device_SWAT_Arsenal.md), failure-mode libraries, and protocol updates maintained by SA.

- **Mode scaling**: Start on Auto/Expert; escalate to Heavy only when residual uncertainty stays high or the topic is high-stakes. Multiple independent runs possible for ensemble coverage.
- **Prompts**: Ready-to-use system prompts in MAD_Agent_Prompts.md.
- **Evolution principle**: Science-like constant review rather than premature consensus. Residual uncertainty is preserved and made explicit.

Design Intent

- Catch incomplete or first-pass AI information before the user has to discover gaps.
- Produce portable “DNA” of adversarial verification that can be applied to any domain (flights, app design, devices, public transport, language tools, etc.).
- Continuously improve via SA rather than freeze a static setup.

This architecture is deliberately lightweight for manual testing while structured enough to scale into automated multi-round or multi-agent ensembles as needed.

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